Practical Assignment – I

Getting Familiarity with C++

1. Write a C++ program to print Hello World.
2. Write a C++ program to display any number entered by the user.
3. Write a C++ program to input three numbers and display its total and average.
4. Write a C++ program to enter Day Temperature of 5 cities of Gujarat. Display average temperature.
5. Write a C++ program to find simple interest.
   SI = (P*R*N)/100 with float values
6. Write a C++ program to calculate area of circle.
   Area = π * r * r
7. Write a C++ program to read two numbers from user and print their addition, subtraction, multiplication and division on screen.
8. Write a C++ program to input marks of five subjects of a student and calculate total marks and percentage.

ASSIGNMENT -II
Getting familiar with C++
Class and Object

1. WAP to print your name, age and city and pin code on screen (Using Class).
2. Write a program to print the area of a rectangle by creating a class named 'Area' having two functions. First function named as 'setDim' takes the length and breadth of the rectangle as parameters and the second function named as 'getArea' returns the area of the rectangle. Length and breadth of the rectangle are entered through keyboard.

3. WAP to display addition, subtraction, multiplication and division of two integers on screen.

   Declare Class Calculation
   Declare data member num1 and num2 in Private section
   Write member function for initialize num1 and num2
   Write member function for each operation.

4. WAP to find area of circle. Area of Circle = PI * r * r Where, PI = 3.14
   (Using Class and Object)

5. Write a C++ program to create a class for student to get and print details of a student. Following are the details of a student:

   Studid, name ,sem, branch

6. Write a C++ program to demonstrate ATM money withdrawal process by taking following private data members:

   Accountno, balance;

   The withdrawal function should return remaining balance to the user and should deduct withdrawal amount from balance. If withdrawal amount > balance print appropriate message on screen (Not enough balance)

   The Deposit function should return updated balance to user.

7. Write a C++ program to create a class for student to get and print details of a student. Following are the details of a student:

   Student_id, Name, Branch,Sub1_mark,Sub2_mark,Sub3_mark,Sub4_mark,Sub5_mark

   Write member function to calculate Percentage, Class (Dist,First,Second,Pass) of student

8. C++ program to create a class for Employee to get and display following Employee information:

   Empcode, Basicsalary

   Write member function to calculate Net salary
   DA=174% of Basic salary
   HRA=10% of Basic salary
   TA=500
   Income tax=5 % of basic salary, if basic salary >50000
   Income tax=0, if Basic salary <=50000

   Netsalary=(basicsalary+da+hra+ta) - income tax

9. Do above program for 5 number of Employees.(Using Array of Object)

10. Write a program to read time in hh:mm:ss and display answer in only seconds. For example if user enters 2:15:30 then it should display 8130 seconds.
    Input:
    Enter Hours:2
    Enter minutes:15
    Enter Seconds:30
    Output:
    8130 seconds

ASSIGNMENT -III
Getting familiar with C++
Constructor

1. Write a program to print the area of a rectangle by creating a class named 'Area' having one function. Length and breadth of the rectangle are entered through keyboard using Parameterized constructor.

2. WAP to display addition, subtraction, multiplication and division of two integers on screen.

   Declare Class Calculation

   Write Parameterized constructor for initialize num1 and num2

   Write member function for each operation.

3. Write a C++ program to demonstrate ATM money withdrawal and deposit process by taking following private data members:

   Accountno, balance;

   Account no and balance data member initialize using parameterized constructor

   Write three function 1. Deposit 2. Withdraw 3. Balance

   Write menu driven choice

   1. Deposit
   2. Withdraw
   3. Balance
   4. Exit

   Program stop execution when user enter choice 4.

4. Create a class “Mobile” with attributes: brand, price, color, width, height. Use constructor to set default values of these attributes. Write function to display details of all attributes.

5. Create a class “Mobile” with attributes: brand, price, color. Enter detail of five different mobile. (Using Array of object).

Display total number of mobile having price greater than 5000.

Display Brand, Price and color for all mobiles for price range 1000 to 10000

6. C++ program to create a class for Employee to get and display following Employee information:

    Empcode,Emp name, Basicsalary

Count the created objects using static member.

7. Create a class “Student” having following data members:

    studid, name, marks (of 5 subject), percentage.

    Use getdata and show functions to input and display student data.
    Create private function to calculate percentage.
    Display detail of student who secured highest percentage.
    Create N number of student using array of object.

8. Create a class Student with Rollno,mark1,mark2,mark3 data member and write appropriate function to setdata .Write another class name sport with sport_mark data member and set sport_mark. Find Resul(mark1+mark2+mark3+sport_mark) of student using Friend function

ASSIGNMENT IV
Getting familiar with Inheritance in C++
Inheritance

1. Write a program that defines a shape class with a constructor that gives value to width and height. Then define two sub-classes triangle and rectangle, that calculate the area of the shape. In the main, define two objects a triangle and a rectangle and then call the area () function.

2. Write a program with a mother class animal. Inside it define a name and an age variables, and set_value () function. Then create two sub class Zebra and Dolphin which write a message telling the age and name of animal, also giving some extra information for both sub class (e.g. place of origin).place of origin of zebra is Earth and place of origin of dolphin is water.

3. Write a program as per following details

Create one base class Teacher

Data members Name, Department, College name, Email id.

Create sub classes for Math Teacher, English Teacher, and Science Teacher

Data member Qualification, Expertise and salary.

Display following details for each teacher

Name:

Department:

College name:

Email id:

Qualification:

Expertise:

Salary:

4. Write a program as per following details

Create one base class DRUG with following data members

Category- (i.e. **stimulants, inhalants, cannabinoids)**

Date_of_manufacture, Company name

Create one sub class TABLET derived from DRUG with following data members

Tablet name, Price

Create one sub class PainReliever derived from TABLET with data member

Dosage_units: i.e(1 or 2 or 3)
Side_effects : i.e (Nausea, Drowsiness, Dizziness.)

Use_within_days: i.e(10 or 20 or 30).

Use appropriate member function for setting and Getting above details and display details in main function.

5. Write a program as per following details

Create one base class HOTEL with following data members

Hotel_name,

Hotel_type i.e(Three star,five star)

City

Hotel_rate i.e(2000,3000,5000)

Create one base class FLIGHT with following data members

Flight_no

Source city

Destination city

Seat no

Create one sub class PASSENGER derived from HOTEL and FLIGHT with following data members

Name, Age, city

Write appropriate member functions in each class and display all information in main

6. Write a program as per following details

Create one base class PERSON with following data members

Name, College name

Create one sub class STUDENT derived from PERSON with following data members

Student_id , Marks of five subject, percentage

Member function:

showResult( )-Calculate total,percentage and finding class(Dist,First,second,pass)

Create one sub class EMPLOYEE derived from PERSON with following data members

Emp_id, qualification , basic salary

Member function to calculate Net salary and print Net salary

DA=189% of Basic salary
HRA=10% of Basic salary
TA=500
Income tax=5 % of basic salary, if basic salary >50000
Income tax=0, if Basic salary <=50000

Netsalary=(basicsalary+da+hra+ta) - income tax

Write appropriate setter function in each class and display detail of student and employee in main.

ASSIGNMENT V
Getting familiar with Operator Overloading

1. Write a program to overload unary minus operator

   Ex. Input : obj(10,-20,30) Output : -10,20,-30 (-obj)

2. Write a program to overload pre decrement operator (obj1= --obj)

3. Write a program to overload post decrement operator (obj1=obj--)

4. Write a program to overload Binary *(multiply) operator for object of same class

Ex. class student s1,s2,s3 s3=s1*s2

5. Write a program to overload Binary * (multiply) operator for object of different class

Ex. Class unit_test, class Practical Practical result=object of unit_test+object of practical

6. Write program to overload = = operator to compare two object of student class

```
Class student
{
        Int mark;
        Public:
       Student (int tmark)
       {
        mark=tmark;
       }
};

Int main() {

Student s1(100),student s2(200),student s3(100);

If(s1==s2)

   Cout<<" no same marks"

If(s1==s3)

  Cout<< "same marks"
```

ASSIGNMENT VI
Working with File

1.Write a program to create a file “Product.txt” and write data as Product Name, Product Price and Batch Number in a file.

2. Write a program to append Product Type and Product Weight in a file “Product.txt”.

3. Write a program to create a file “Patient.txt” and enter Patient ID, Patient Name, Disease Name and City in a file. Now read patient details and display all the details on a terminal.

4. Write a program to append Doctor Name and Doctor Speciality in “Patient.txt” file and display the data on a console.

5.Create a class “Mobile” with attributes: brand, price, color. Use getData() values of these attributes and store object in file. Write function displayData() to display object from file.

6. Create a class “Student” having following data members:

studid,name,marks (of N subject), percentage. Create private function to calculate percentage. Use getData() functions get data from keyboard and store data to the file. Write function displayData() to display object from file.

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Smt. Chandaben Mohanbhai Patel Institute of Computer Applications
BCA – Semester IV
CA221 – Data Structures and Algorithms

Practical Set – I Programs on Array and Linked List	
1	Write a program to accept n elements and print it.
2	Write a program to accept n elements and find the summation and average of it.
3	Write a program to accept n elements and print all odd elements and count odd elements.
4	Write a program to accept n elements and find the maximum and minimum element from it.
5	Write a program to perform the following operations on a stack. (Implement the stack using array and linked list both) • PUSH • POP • ISEMPTY • ISFULL • PEEP
6	Write a program to perform the following operation on a simple queue. (Implement the queue using array) • Insert an element • Remove an element.
7	Write a program to perform the following operation on a circular queue. (implement the queue using array) • Insert an element • Remove an element
8	Write a program to create a Singly Linked List in LIFO fashion.
9	Write a program to create a Singly Linked List in FIFO fashion.
10	Write a program to create a sorted Singly Linked List.
11	Write program perform the following operations on a Singly Linked List. 1. Insert an element 2. Delete an element 3. Find the sum of elements of the List

	4. Count number of the nodes in the Linked List 5. Search a given element in the Linked List. 6. Reverse the Linked List. 7. Make a copy of the given Linked List 8. Concatenate two Linked List 9. Merge two Linked List. 10. Find the union of the two given Linked List 11. Find the intersection of the two given Linked List.
12	Write a program to create a sorted Doubly Linked List.
13	Write a program to create a Doubly Linked List in LIFO fashion.
14	Write a program to create a Doubly Linked List in FIFO fashion.
15	Write a program perform the following operations on a doubly Linked List. 1. Insert an element 2. Delete an element 3. Find the sum of element of the List 4. Count number of the nodes in the Linked List 5. Search a given element in the Linked List. 6. Reverse the Linked List 7. Make a copy of the given Linked List 8. Concatenate two Linked listed. 9. Merge two Linked List. 10. Find the union of the two given Linked List. 11. Find the intersection of the two given Linked List.
16	Write a program to create a Circular Single Linked List.
17	Write a program to create a Double Linked List.

Practical Set – II Programs on Binary Tree	
1	Write a program to create a binary search tree.
2	Write a program to create a binary search tree and print it's elements in inorder. (write iterative code)
3	Write a program to create a binary search tree and print its elements in preorder. (write iterative code)
4	Write a program to create a binary search tree and print its elements in postorder. (write iterative code)
5	Write a program to make another copy of a given binary search tree.
6	Write a program to count no of leaf nodes in a given binary tree.
7	Write a program to search an element in a given binary search tree.
8	Write a program to accept n elements and check particular element is present how many times. 1. Linear Search 2. Binary Search
9	Write a program to accept n elements and arrange all the elements in Ascending order. 1. Bubble Sort 2. Selection Sort 3. Insertion Sort 4. Shell Sort 5. Quick Sort 6. Merge Sort 7. Radix Sort 8. Heap Sort 9. Topological Sort

ASSIGNMENT-1

Objectives

- **Hands on with Basics of Data Definition Language (DDL) commands Like Create Table, Alter Table, Truncate Table, Drop table etc.**
- **Hands on with Basics of Data Manipulation Language (DML).**
- **To understand and apply constraints to the tables.**

Create Following tables

1. Table1 : stud_master

Column Name	Datatype and Size	Constraints	Description
Sid	varchar(10)	Primary Key. Should contain BCA or MCA	Student ID. Example:21BCA001
Fname	Varchar2(50)		Name of a student. Example: Amit
Lname	Varchar2(50)		Surname of a student. Example: Patel
Branch	varchar2(10)		
Semester	Char(1)		
City	Varchar2(30)		
DOB	DATE		Format: DD-MMM-YYYY. For example: 18-JAN-2003
Gender	CHAR(1)	Should Contain M or m or F or f	

Insert following data in above table:

Sid	Fname	Lname	Branch	Semester	City	DOB	Gender
21BCA001	Vishal	Patel	BCA	2	Anand	18-APR-2002	M
21BCA002	Mahesh	Mandavia	BCA	2		20-JUL-2001	M
20MCA003	Monesh	Viradia	MCA	2	Surat	21-MAR-1998	M
19BCA024	Frida	Pinto	BCA	4	Vasad	18-APR-2002	F
20BCA027	Tanvi	Sharma	BCA	2	Nadiad	04-JAN-2002	F
18BCA045	Vishal	Shah	BCA	4	Anand	08-MAR-2002	M
17MCA060	Shailesh	Macwan	MCA	4		15-APR-2001	M
19BCA100	Rupesh	Misra	BCA	4	Surat	25-AUG-2002	M
17MCA002	Vishal	Mehta	MCA	4	Nadiad	18-OCT-2002	M
17BCA124	Priya	Dave	BCA	4	Vasad	06-FEB-2002	F

2. Table 2: sub_master

Column Name	Datatype and Size	Constraints	Description
Subcode	varchar(10)	Primary Key. Should start with CA	Example:CA119
Subject	Varchar2(40)		subject name
Th_credit	Number(1)	Should be > 0 and <7	Credits in theory portion of the subject

Pr_credit	Number(1)	Should be > 0 and <4	Credits in practical portion of the subject

Insert following data in above table:

Subcode	Subject	Th_credit	Pr_credit
CA119	Database Fundamentals	4	3
CA118	Advanced Programming	4	3
CA117	Operating system concepts	4	3
CA221	Data Structures and Algorithms	4	3
CA321	Basics of Mobile Applications		3
CA115	Management Information System	3	0
CA836	Framework and Applications	0	3
CA835	Artificial Intelligence	3	0

3. Table3 : stud_subject

Column Name	Datatype and Size	Constraints	Description
Sid	varchar(10)	Should contain BCA or MCA	Student ID. Example:21BCA001
Subcode	varchar(10)	Should start with CA	Example:CA119

Insert following data in above table:

Sid	Subcode
21BCA001	CA119
21BCA002	CA119
20MCA003	CA119
19BCA024	CA321
20BCA027	CA221
18BCA045	CA836
17MCA060	CA836
19BCA100	CA835
17MCA002	CA835
17BCA124	CA119
21BCA001	CA119
21BCA002	CA119
20MCA003	CA321

Solve Following queries:

1. Display all student details.
2. Display all subject details.
3. List all the students from Anand city.
4. List all the female students.
5. List all the male students.
6. Display name of the students born in April as per the following format:

Name	DOB
Vishal Patel	18-Apr-2002
Shailesh Macwan	15-Apr-2001

7. List students of BCA branch.
8. List students of MCA branch.
9. List students who have taken admission in year 2017.
10. List BCA students who have taken admission in year 2018.
11. Display details of student with no city.
12. Display gender wise total number of students (Total Male and Total Female).
13. Display details of all students of semester 2.
14. Display age of all the students in following format:

Name	**AGE**
Vishal Patel	16
Shailesh Macwan	17

15. Display all female students of BCA.
16. Display all male students of MCA.
17. Display all distinct branches
18. Display total credits in all subject as per following format:

Subcode	Subject	Total Credits
CA119	Database Fundamentals	7
CA118	Advanced Programming	7
----	------	-------

19. Display details of Artificial Intelligence subject.
20. Display only theory subjects.
21. Display only practical subjects.
22. Display all studid and subcode combinations.
23. Display studids who are enrolled for subject code CA836.
24. Display all subject codes of studid 17BCA060
25. Display total number of distinct subject codes.
26. Display total number of distinct studids.
27. Display list of students who have not done course registration.
28. Change the city of roll no 21BCA001 to Mumbai.
29. Insert the city of roll no 21BCA002
30. Increase the size of branch column to varchar2(20).
31. Delete the subject ‘Framework and Applications’.

<u>Queries of Expressions/Functions</u>

1. Find eldest student. HINT: use max function.
2. Find youngest student. HINT: use min function.
3. Find total number of students. HINT: use count function.
4. Find total credits in all theory subjects. HINT: use sum function.
5. Find total credits in all practical subjects. HINT: use sum function.
6. Display student’s DOB in following format: HINT: use to_char() function.

Name	DOB
Vishal Patel	April 18, 2002
Shailesh Macwan	April 15, 2001

7. Display student's DOB in following format: HINT: use add_months() function.

Name	DOB
Vishal Patel	18-Oct-2002
Shailesh Macwan	15-Oct-2001

8. Find difference in months between your DOB and today's date.
9. Add following column in the **stud_master created.**

Mobile	Number(10)		

10. Create following temporary table. Name: **Stud_temp**

Column Name	Datatype and Size	Constraints	Description
Temp_ID	varchar(10)	Not Null	Student ID. Example:21BCA001
Remarks	varchar(20)		

    1. Enter the Data in **Stud_temp** table.
    2. Display content of the **Stud_temp** Table.
    3. Truncate the **Stud_temp table** without explicitly deleting the rows.
    4. Again Display content of the **Stud_temp** Table.
    5. Drop The Table Stud_temp.

ASSIGNMENT-2

Objectives

- **Hands on with Basics of Data Definition Language (DDL)**
- **To apply JOINS, GROUPBY and Advance SQL queries.**

(i) Create tables according to the following definition.

CREATE TABLE DEPOSIT (ACTNO VARCHAR2(5) ,CNAME VARCHAR2(18) , BNAME VARCHAR2(18) , AMOUNT NUMBER(8,2) ,ADATE DATE);

CREATE TABLE BRANCH(BNAME VARCHAR2(18),CITY VARCHAR2(18));

CREATE TABLE CUSTOMERS(CNAME VARCHAR2(19) ,CITY VARCHAR2(18));

CREATE TABLE BORROW(LOANNO VARCHAR2(5), CNAME VARCHAR2(18), BNAME VARCHAR2(18), AMOUNT NUMBER (8,2));

(ii) Insert the data as shown below.

DEPOSIT

ACTNO	CNAME	BNAME	AMOUNT	ADATE
100	ANIL	VRCE	1000.00	1-MAR-95
101	SUNIL	AJNI	5000.00	4-JAN-96
102	MEHUL	KAROLBAGH	3500.00	17-NOV-95
104	MADHURI	CHANDI	1200.00	17-DEC-95
105	PRMOD	M.G.ROAD	3000.00	27-MAR-96
106	SANDIP	ANDHERI	2000.00	31-MAR-96
107	SHIVANI	VIRAR	1000.00	5-SEP-95
108	KRANTI	NEHRU PLACE	5000.00	2-JUL-95
109	MINU	POWAI	7000.00	10-AUG-95

BRANCH

VRCE	NAGPUR
AJNI	NAGPUR
KAROLBAGH	DELHI
CHANDI	DELHI
DHARAMPETH	NAGPUR
M.G.ROAD	BANGLORE
ANDHERI	BOMBAY
VIRAR	BOMBAY
NEHRU PLACE	DELHI
POWAI	BOMBAY

CUSTOMERS

ANIL	CALCUTTA
SUNIL	DELHI
MEHUL	BARODA
MANDAR	PATNA

MADHURI	NAGPUR
PRAMOD	NAGPUR
SANDIP	SURAT
SHIVANI	BOMBAY
KRANTI	BOMBAY
NAREN	BOMBAY

BORROW

LOANNO	CNAME	BNAME	AMOUNT
201	ANIL	VRCE	1000.00
206	MEHUL	AJNI	5000.00
311	SUNIL	DHARAMPETH	3000.00
321	MADHURI	ANDHERI	2000.00
375	PRMOD	VIRAR	8000.00
481	KRANTI	NEHRU PLACE	3000.00

From the above given tables perform the following queries:
(1) Describe deposit, branch.
(2) Describe borrow, customers.
(3) List all data from table DEPOSIT.
(4) List all data from table BORROW.
(5) List all data from table CUSTOMERS.
(6) List all data from table BRANCH.
(7) Give account no and amount of depositors.
(8) Give name of depositors having amount greater than 4000.
(9) Give name of customers who opened account after date '1-12-96'.

2. **Create table and insert sample data in tables**.

Create Table Job (job_id, job_title, min_sal, max_sal)

COLUMN NAME	DATA TYPE
job_id	Varchar2(15)
job_title	Varchar2(30)
min_sal	Number(7,2)
max_sal	Number(7,2)

Create table Employee (emp_no, emp_name, emp_sal, emp_comm, dept_no)

COLUMN NAME	DATA TYPE
emp_no	Number(3)
emp_name	Varchar2(30)
emp_sal	Number(8,2)
emp_comm	Number(6,1)
dept_no	Number(3)

Create table deposit(a_no,cname,bname,amount,a_date).

COLUMN NAME	DATA TYPE

a_no	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Number(7,2)
a_date	Date

Create table borrow(loanno,cname,bname,amount).

COLUMN NAME	DATA TYPE
loanno	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Varchar2(7,2)

Insert following values in the table **Employee**.

emp_n	emp_name	emp_sal	emp_comm	dept _no
101	Smith	800		20
102	Snehal	1600	300	25
103	Adama	1100	0	20
104	Aman	3000		15
105	Anita	5000	50,000	10
106	Sneha	2450	24,500	10
107	Anamika	2975		30

Insert following values in the table **job**.

job_id	job_name	min_sal	max_sal
IT_PROG	Programmer	4000	10000
MK_MGR	Marketing manager	9000	15000
FI_MGR	Finance manager	8200	12000
FI_ACC	Account	4200	9000
LEC	Lecturer	6000	17000
COMP_OP	Computer Operator	1500	3000

Insert following values in the table **deposit**.

A_no	cname	Bname	Amount	date
101	Anil	andheri	7000	01-jan-06
102	sunil	virar	5000	15-jul-06
103	jay	villeparle	6500	12-mar-06
104	vijay	andheri	8000	17-sep-06

105	keyur	dadar	7500	19-nov-06
106	mayur	borivali	5500	21-dec-06

3. Displaying data from Multiple Tables (join)

(1) Give details of customers ANIL.
(2) Give name of customer who are borrowers and depositors and having living city nagpur
(3) Give city as their city name of customers having same living branch.
(4) Write a query to display the last name, department number, and department name for all employees.
(5) Create a unique listing of all jobs that are in department 30. Include the location of the department in the output
(6) Write a query to display the employee name, department number, and department name for all employees who work in NEW YORK.
(7) Display the employee last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.
(8) Create a query to display the name and hire date of any employee hired after employee SCOTT.

4. To apply the concept of Aggregating Data using Group functions.

(1) List total deposit of customer having account date after 1-jan-96.
(2) List total deposit of customers living in city Nagpur.
(3) List maximum deposit of customers living in bombay.
(4) Display the highest, lowest, sum, and average salary of all employees. Label the columns Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.
(5) Write a query that displays the difference between the highest and lowest salaries. Label the column DIFFERENCE.
(6) Create a query that will display the total number of employees and, of that total, the number of employees hired in 1995, 1996, 1997, and 1998
(7) Find the average salaries for each department without displaying the respective department numbers.
(8) Write a query to display the total salary being paid to each job title, within each department.
(9) Find the average salaries > 2000 for each department without displaying the respective department numbers.
(10) Display the job and total salary for each job with a total salary amount exceeding 3000, in which excludes president and sorts the list by the total salary.
(11) List the branches having sum of deposit more than 5000 and located in city bombay.

5. To solve queries using the concept of sub query.

(1) Write a query to display the last name and hire date of any employee in the same department as SCOTT. Exclude SCOTT
(2) Give name of customers who are depositors having same branch city of mr. sunil.
(3) Give deposit details and loan details of customer in same city where pramod is living.
(4) Create a query to display the employee numbers and last names of all employees who earn more than the average salary. Sort the results in ascending order of salary.
(5) Give names of depositors having same living city as mr. anil and having deposit amount greater than 2000
(6) Display the last name and salary of every employee who reports to ford.
(7) Display the department number, name, and job for every employee in the Accounting department.
(8) List the name of branch having highest number of depositors.
(9) Give the name of cities where in which the maximum numbers of branches are located.

(10) Give name of customers living in same city where maximum depositors are located.

6. Manipulating Data

(1) Give 10% interest to all depositors.
(2) Give 10% interest to all depositors having branch vrce
(3) Give 10% interest to all depositors living in nagpur and having branch city bombay.
(4) Write a query which changes the department number of all employees with empno 7788’s job to employee 7844’current department number.
(5) Transfer 10 Rs from account of anil to sunil if both are having same branch.
(6) Give 100 Rs more to all depositors if they are maximum depositors in their respective branch.
(7) Delete depositors of branches having number of customers between 1 to 3.
(8) Delete deposit of vijay.
(9) Delete borrower of branches having average loan less than 1000.

ASSIGNMENT-3

Objective:

- **To describe constant, variable, variable initialization, and basic operations on the variables in PL/SQL.**

1. PL/SQL Block

1 Write a PL/SQL block to print 'WELCOME' message.

2 Write a PL/SQL block to input and to print your personal detail in the following format using appropriate variable.

Name: <<Name>>
Address: <<Address>>
City: <<City>>
Mobile: <<Mobile Number>>
Birth Date: <<Birth date>>

3 Write a PL/SQL block to print to initialize two number variables and print the total of it.

4 Write a PL/SQL block to swap the values of two number type of variables. Print the values of variables before and after the swap.

5 Write a PL/SQL block to print minimum of two numbers.

6 WRITE A PL/SQL BLOCK TO FIND MAX OF THREE NUMBERS.

7 WRITE A PL/SQL BLOCK TO PRINT GRADE OF THE STUDENT
INPUT : ROLLNO, MARKS1, MARKS2, MARKS3
CALCULATE TOTAL = MARKS1 + MARKS2 + MARKS3 / 3
BASED ON TOTAL CALCULATE PER
GRADE IS CALCULATED AS PER FORMULA
IF PER >80 GRADE IS AA
IF PER BETWEEN 70 AND 80 GRADE IS A
IF PER BETWEEN 60 TO 70 GRADE IS AB
IF PER BETWEEN 50 TO 60 GRADE IS B
IF PER BETWEEN 40 TO 50 GRADE IS C
IF PER LESS THAN 40 GRADE IS FAIL

8 WRITE A PL/SQL BLOCK TO PRINT 10 NUMBERS USING WHILE AND FOR LOOP

9 WRITE A PL/SQL BLOCK TO PRINT MULTIPLICATION TABLE USING WHILE AND FOR LOOP

10 WRITE A PL/SQL BLOCK TO PRINT FACTORIAL OF GIVEN NUMBER.

ASSIGNMENT-4

Objective:

- **To describe discuss DML statement in PL/SQL with exception handling also.**

PL/SQL Implicit Cursor

1 Write a PL/SQL block to display name and mobile number from EMP table whose id is 101.

2 Write a PL/SQL block to insert 5 numbers in temp table, update the any number by taking input from user, delete any number as per user choice. Also print appropriate message.

3 WRITE A PL/SQL BLOCK TO INSERT ODD NUMBERS IN T1 TABLE AND EVEN NUMBERS IN T2 TABLE
WRITE A PL/SQL BLOCK TO DELETE VALUES FROM TABLE T1
WRITE A PL/SQL BLOCK TO DELETE LAST 50 RECORD OUT OF 100 FROM TEMP TABLE

4 Create table of patient with fields
PATIENT(pid integer, pnm varchar2(10), age integer)
---> TAKE PATIENT ID AS PRIMARY KEY,
AGE SHOULD BE GREATER THAN ZERO
ADD GENDER FIELD INTO PATIENT TABLE AND UPDATE THE RECORD

Write a PL/SQL block which will print
the patient report in the following format
INPUT PATIENT ID FROM USER AND
COMPARE IT WITH TABLE ID AND DISPLAY ACCORDINGLY

patient id: XXX Patient name: XXX
AGE <20 : PRINT IT
AGE >= 20 : PRINT IT
AGE >20 and AGE <=30 : PRINT IT
AGE > 30 & AGE <=40 : PRINT IT
AGE >40 & AGE <60 : PRINT IT
AGE >=60 : DISPLAY IT

5 WRITE A PL/SQL BLOCK TO DO THE FOLLOWING:
INPUT 20 NUMBERS FROM USER
A) FIRST ADD ALL NUMBERS IN T3 TABLE
B) IF NUMBER IS 5,10,15 THEN DO UPDATE T3 (ADD 20 IN)
ELSE DELETE OTHER RECORDS
Print appropriate message for valid and invalid record.

ASSIGNMENT-5

Objective:

- **To describe SELECT-into clause – only one output statement in PL/SQL**

```
CREATE TABLE DOCTOR(DID INTEGER PRIMARY KEY, DNAME
VARCHAR2(10))

CREATE TABLE PATIENT(pid integer PRIMARY KEY,
    patientnm varchar2(10),
    age integer CHECK(AGE > 0),
    GENDER CHAR(1) CHECK(GENDER IN ('M','F','m','f')),
     DID INTEGER REFERENCES DOCTOR)

INSERT 5-7 RECORDS IN BOTH TABLE

SELECT * FROM DOCTOR
SELECT * FROM PATIENT
```

1. Write a PL/SQL block to print senior citizen(maximum age of patient).

2. Write a PL/SQL block to print name of patient who is youngest.

3. WRITE A PL/SQL BLOCK TO COUNT TOTAL NUMBER OF RECORDS FROM PATIENT TABLE

4. WRITE A PL/SQL BLOCK TO DISPLAY MESSAGE OF PATIENT
   E.G. TAKE INPUT AS PID, EXTRACT AGE AND DO THE FOLLOWING

   TAKE AGE FROM TABLE AND BASED ON CONDITION DISPLAY THE RECORD
   IF AGE >= 60 THEN DISPLAY- YOU ARE SENIOR CITIZEN
   IF AGE >40 AND AGE <60 DISPLAY- YOU ARE MATURE CITIZEN
   IF AGE >20 AND AGE <=40 DISPLAY- YOU ARE YOUTH CITIZEN
   IF AGE >14 AND AGE <= 20 DISPLAY - YOU ARE TEENAGER
   IF AGE <=14 DISPLAY YOU ARE CHILD

5 WRITE A PL/SQL BLOCK TO PRINT DEPT NAME OF DEPT ALONG WITH EMPLOYEE NAME WHOSE DEPTNO = 2

ASSIGNMENT-6

Objective:

- **To describe Explicit cursor- ALL records from table or selected records from table and manipulate data based on logic.**

1 Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

Write a PL/SQL block to display all the records.
(use explicit cursor and cursor-For loop)

2 Write a PL/SQL block to display name, phone-no and designation of female employees.

3 Write a PL/SQL block to display highest paid first 5 employees.

4 ADD bonus field in EMP table.

Write a PL/SQL block to calculate bonus based on condition. After calculation of bonus of all the employees, display on screen as well as update it.
IF salary greater than 50k then bonus is 10% of salary
IF salary greater than 25k and less than 50k bonus is 15% of salary
If salary less than 25k then bonus is 20% of salary.

5 Assume that you have EMP and DEPT table with proper records:

Find out the name of employee who has experienced in the Electrical department more than 5 years.

6 Assume that you have EMP and DEPT table with proper records:
Write a PL-SQL block that will print the following report:

Id	NAme	Gender	Phone-no	Designation	Deptid	Deptname
1	Mahendra	M	9999222234	Manager	101	MCA
2	Virat	M	9873627789	Sales Manager	102	Civil
3	Rohit	M	8762349087	Clerk	101	MCA
4	Renu	F	7896734567	Accountant	103	Electrical

Total Records Printed : 4

ASSIGNMENT-7

Objective:

- **To discuss Stored Procedure in PL/SQL from block or direct.**

Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

1 Write a Procedure which display average salary of employee.

2 Write a Procedure which pass two parameters, calculate sum of two values and display it in procedure.

3 Write a Procedure which will display total no. of employees whose salary greater than 10000 and execute it.

4 Write a Procedure which will display mobile number of employee whose id is given by user. (pass as parameter)

5 Create a stored procedure that takes and input parameter as department- id and generate an output from the procedure as number of employees who are working in that department.

6 Write a procedure for following task:
When new employee joins in the institute, check if empid is present in the institute or not and the no. of already registered employees *not e*xceeding the max capacity for that institute**.**

7 Write a procedure to display top five highest paid employees whose designation is Officer.

8 Write a procedure to display name of employee from emp table in a given format:
(first name, middle name, last name), use CASE statement to have 4 options.
For example,

- Display name in the format “ RLP “
- Display name in the format “ R.L.Patel “
- Display name in the format “ Rambhai L. Patel “
- Display name in the format “ Rambhai Laxmanbhai Patel”

9 Enter the employee id from user. Check whether the name exists in table for corresponding id. If yes, Write a PL/SQL procedure to find total no of vowels from employee name. If no, display appropriate message.

10 To vote for Election, write a Procedure which will display the following details:
Eligible to vote(>18 years) || Not Eligible (less than 18 years)
Id & name of employee || id & Name of Employee

ASSIGNMENT-8

Objective:

- **To discuss Stored Function and local function in PL/SQL from block.**

1. Write a function to take two numbers from user and print summation of it.
2. Write a function whether given number is odd or even.
3. Write a function that returns total number of incomplete jobs, using table JOB(jobid, type_of_job, status)
4. Write a function to which a birthdate is passed and it returns the age in terms of years as of today and a procedure/block which calls this function to display.
5. Write a function which displays the number of items whose weight fall between a given range for a particular color using table ITEM(itemno, name, color, weight)
6. Create a function to return count of all employees who works in MCA department. This function is called in procedure which displays the output in proper format.
7. ISSUE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE)

   After calculating the fine store the required information in the FINE table so that a report can later be printed out.

   FINE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE, TOT_DAYS, FINE)

   Write a procedure or function for the given a table, check which students have to pay fine and the amount of fine to be paid. A student can keep the book for 15 days. If the number of days has exceeded 15 then, the fine is calculated as follows:

   Up to 7 days : 50 paise per day

   8 – 15 days : Re 1 per day (to be counted from the 8th day onwards)

   More than 15 : Re 1.5 per day (to be counted from the 16th day onwards)

8. NEWS_PAPER(news_id, name_of_paper, language, month_frequency, price)

   Write a procedure that will display name of paper which is of lowest price.
   Write a function which will count total number of newspaper exists language wise.
   (English language : 999 papers
   Gujarati language : 999 papers)
   Hindi language : 999 papers)

ASSIGNMENT-9

Objective:

- **To discuss Database Trigger in PL/SQL.**

Assume above EMP and DEPT table exists with records.

1 Write a code which demonstrate simple trigger.

2 Create a trigger on EMP table that tracks the UPDATE on it. Each record being updated on it shall be tracked and kept as backup in EMP_TRAN table

3 Create a trigger on EMP table that tracks the DELETE on it. Every record being deleted from it shall be kept in EMP_TRAN_BACKUP table.

4 Create a trigger on EMP table that tracks UPDATE being performed on SAL column of the table. If the UPDATE is being performed between 2.30 pm to 3.30 pm then copy the effect on EMP_UPDATE_TRACK table

5 Write a trigger which restrict to enter or update the transaction on Saturday or Sunday
OR
Write a PL\SQL trigger which will not allow to insert records on saturday and sunday.

6 Write a trigger which restrict user to enter name in EMP table either starting with 'X' or is less than 2 letters

7 ADD Blood group in EMP table.

Write a trigger which only accept blood_gr in range(with +ve and –ve blood gr) of A ,B, AB, O.

8 Write a trigger in which accepts empid from user and check whether sal>65000 , don't allow to insert or update emp table.

9 Create a trigger which ensures that Emp name that to be inserted is in capital.

10 Write a trigger which deletes every record from result table if student table is empty.
student(studno,fnm,mnm,lnm,degree,bdth);
result(studno,math,eng ,ss,sci)

11 Write a trigger which accepts student records whose age is between 18 to 30.

12 Consider the table: (Eno, Ename, Basic, Hra, Da, Total)
Assume the following :
* Hra is given in Rupees
* DA is given as percentage as basic
* Total salary will be Basic + da*basic/100 + hra
(i) Write a trigger so that the record on insertion contains basic salary in the range of 5000 to 18000;
DA in the range 16% to 32%; and HRA not more than 3000.

ASSIGNMENT-10

Objective:

- **To discuss PL/SQL Parameterized cursor, Exceptional handling and package.**

1 Item (Item code, item name, unit price)
Bill (bill number, bill date)
Bill_item (bill number, item code, number of units purchased)

Do the following:

(a) For a given bill number, print the bill in the following format.

Bill no.: Bill Date:

Sr. No.	Item code	Item name	Units purchased	unit price	total price
	###	xxxxx	xxxxxxxxxxxxxxxxx #####	####.##	#######.##

Bill Amount: #########.##

(b) Add one more attribute named bill_amount in the Bill table. Update the Bill table to store bill amount.

2 **PATIENT(pid, pname, age, gender, doctor-id)**
Prepare following report for a Doctor information related to total patient age wise in his hospital.

Gender	Patient in Age Group					Total
	1-12	13-25	25-50	50-100	Above 100	
----	----					----
Total	-------	--------	------	-------	--------	-----

Patient (pid, pame, age, gender , doctor-id)
Appoint(pcode, app_date, time , doctor_id)
Doctor(doctor_id, name, specialization)

Prepare a report from above table taking date, shift, , doctor_id as parameters.
Shift is Morning if time are from 9.00 Am to 1.00 pm and
Shift is Evening if time are from 5.00 to 9.30 pm.

APPOINTMENT SCHEDULE

Date : Doctor name :
Shift : Specialization :

Time	Patient name	Age
9.00		
9.10		
9.20		
…..		

Total no. of patients : ___________

3 ALBUM(album_no, album_name, album_type, release_date, singer_name, no_songs)

Write a procedure which takes parameter as singer name and print the following album details:

Singer name : ______________

Album no	Album name	Type	Release Date	Number of Songs
99	Xx	Xx	Xx	99
99	Xx	Xx	Xx	99

4 Create table and insert proper data.

Tour_Details(tour_no, start_date, group_size, source, destination, vehicle_no, driver_name)

Write a PL/SQL block which will print tour details, a driver is going to take it. (pass driver_no as parameter)

Driver Name : ____________ Vehicle number : __________________

Tour Details

Source	Destination	Group size	Start date	Total days
Xxx	Xxx	999	Xxx	999

5 CUST(cno, accno, name, add, city , phone, email)
ACCOUNT(accno, amount)

Write a stored procedure for banking purpose which perform following Task :

1. Create a new account (insert)
2. Deposit funds (update)
3. Check account balance (view balance)
4. Close the account (delete)

6 Create table and insert proper data.

TRAIN_MAST(train_no, train_name, arrival_time, departure_time, source, destination)

Write a procedure which will print all train details going from Baroda to Bombay
Write a function which will print arrival time and departure time for a given train. (pass train no as a parameter)

---

7 Given hospital schema

Patient(pat_code, pname, birthdate, gender)

Dentist(dcode, dname, qualification, area, city, experience)

Treatment(dcode, pat_code, treatment_date, treat_details)

Write a procedure or Function to do the following queries: (USE Switch case)

1. Count no. of dentist from ‘lalbaug’ area in baroda city.
2. Display details of dentist with ‘MDS’ as qualification.
3. Display details of treatment given to patient no.’P101’.
4. Display total number of patient who came for treatment as on ’12-Feb-2013’.
5. Display area wise list of dentist who has maximum experience.
6. Display details of youngest dentist.
7. Display dentist details along with their patient.
8. Display details of all female dentist.
9. Count total number of male dentist who live in ‘ahmedabad’.
10. Display patient name without duplicate who come for treatment.
11. Display details of patient who has not given any treatment.

8 **Consider the EMPLOYEE schema as :**

EMPMAST(empno, name, pfno, empbasic , deptno, designation)
HOLIDAY(year, month, holidays, weekoff)
EMPTRAN(empno, month, year, presence, loan(amt))
BANK(accno, empno, bank name, bank branch)

Rules : HRA = 15% of basic
DA = 50% of basic
Medical = 100
PF = 8.33%of basic
Salary is given for (attendance + holidays + weekoff) days.

Print salary slip in foll. Format.

SALARY SLIP of Month :_____ Year _______________

Accno : Name of EMP Designation Dept

Bank name : _________________ Branch : _____________ Dt of Join _________

Presence : __________ Absent days : __________ Sal days : ________

EARNINGS :	DEDUCTIONS
BASIC :	LOAN :
DA :	PF :
HRA :	Prof tax : 100
EDICAL : 100	

TOTAL earnings: __________ TOTAL Deductions: __________

NET PAY : ___________

9 Write a code which demonstrate user defined exception.

10 Write a code which demonstrate package in PL/SQL.

================**THERE IS BEGINNING IN THE END !!!**==========================

Practical Set – I
Programs on Array and Linked List

1 Write a program to accept n elements and print it.

2 Write a program to accept n elements and find the summation and average of it.

3 Write a program to accept n elements and print all odd elements and count odd elements.

4 Write a program to accept n elements and find the maximum and minimum element from it.

5 Write a program to perform the following operations on a stack.
(Implement the stack using array and linked list both)

- PUSH
- POP
- ISEMPTY
- ISFULL
- PEEP

6 Write a program to perform the following operation on a simple queue. (Implement the queue using array)

- Insert an element
- Remove an element.

7 Write a program to perform the following operation on a circular queue.
(implement the queue using array)

- Insert an element
- Remove an element

8 Write a program to create a Singly Linked List in LIFO fashion.

9 Write a program to create a Singly Linked List in FIFO fashion.

10 Write a program to create a sorted Singly Linked List.

11 Write program perform the following operations on a Singly Linked List.

1. Insert an element
2. Delete an element
3. Find the sum of elements of the List
4. Count number of the nodes in the Linked List
5. Search a given element in the Linked List.

6. Reverse the Linked List.
7. Make a copy of the given Linked List
8. Concatenate two Linked List
9. Merge two Linked List.
10. Find the union of the two given Linked List
11. Find the intersection of the two given Linked List.

12 Write a program to create a sorted Doubly Linked List.

13 Write a program to create a Doubly Linked List in LIFO fashion.

14 Write a program to create a Doubly Linked List in FIFO fashion.

15 Write a program perform the following operations on a doubly Linked List.

1. Insert an element
2. Delete an element
3. Find the sum of element of the List
4. Count number of the nodes in the Linked List
5. Search a given element in the Linked List.
6. Reverse the Linked List
7. Make a copy of the given Linked List
8. Concatenate two Linked listed.
9. Merge two Linked List.
10. Find the union of the two given Linked List.
11. Find the intersection of the two given Linked List.

16 Write a program to create a Circular Single Linked List.

17 Write a program to create a Double Linked List.

ASSIGNMENT -I
Introduction to Software testing and Checklist preparation

Prepare the checklist for the below programs in the given format, Program Id: Program Name:
Checklist
Table for checklist: 1.Checklist Item 2. Input 3. Expected Result

Q-1 Prepare checklist to calculate Simple Interest.

Q-2 Prepare checklist to accept a number and check whether number is Odd or Even.

Q-3 Prepare checklist to accept three numbers and find maximum and minimum numbers.

Q-4 Prepare checklist to a calculator that input two operands and one operator and perform the following operations ADD, SUBTRACT, MULTIPLY AND DIVIDE.

Q-5 Prepare checklist to calculate Gross and Net Salary of an employee ,based on the following criteria HRA is 3% of basic DA is 175% of basic TA is 400 Rs.
PT is 80 Rs.
PF is 10% of basic
GROSS = BASIC + HRA + DA+ TA
NET = GROSS – PT – PF

Q-6 Prepare checklist to accept marks of 5 subjects (out of 100), calculate total, percentage and Class of a student based on the following criteria.

Percentage	**Grade**
Above or Equal to 70	A
Between 69 and 60	B
Between 59 and 50	C
Below 50	D

If marks in any subject are below 35 then the will be F.

Q-7 Prepare checklist to a program that read a number between 1 to 7 (including both) and then print corresponding day name from the week. For example, if entered number is 1, then print “Today is Monday”, if entered number is 2, then print “Today is Tuesday”, and so on.

<u>: User story:</u>

As a role of supervisor in a company I can keep track of order details only after successful log in which will ask for username, password. Logging into the system requires OTP verification via SMS.

Prepare checklist for above-mentioned description.

1. Create checklist for following description.

Visit https://www.swaraajsports.com/events.html URL.
Click on Quick pay.

Test case ID	Description	Expected Output	Actual Output

ASSIGNMENT -II
Control Flow Testing

For all the below examples perform the following tasks.

- **Draw the control flow graph for the program.**
- **Calculate the cyclomatic complexity of the program.**
- **List all the independent paths**
- **Design test cases from independent paths.**

Test case ID	Input	Expected Result	Independent Path

1 Write a Program to find a given number is prime number or not.
2 Write a Program to find the factorial of a given number.
3 Write a Program to search a given number from an array. If number is found then give the index of the number if number is not found return -1.
4 Write a Program to find the average values from a given array falling in a given range specified by MIN and MAX.
5 Write a Program to find total number of digits, alphabets and special characters from a given string.
@Testing1523$#

Total digits:4
Total alphabets:7
Total Special characters:3

ASSIGNMENT -III
Getting familiar with Junit using NetBeans

1. Write a Program to find a given number is prime number or not.
2. Write a Program to find the factorial of a given number.
3. Write a Program to search a given number from an array. If number is found then give the index of the number if number is not found return -1.
4. Write a Program to find the average values from a given array falling in a given range specified by MIN and MAX.
5. Write a Program to find total number of digits, alphabets and special characters from a given string.

   @Testing1523$#

   Total digits:4
   Total alphabets:7
   Total Special characters:3

ASSIGNMENT IV
Testing WebApps with Selenium WebDriver using Chrome Driver